

Drew Steindl

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EDUCATION

Yale University <i>Ph.D. in Computer Science</i>	August 2025 – Present New Haven, CT
Vassar College <i>B.A. in Mathematics, GPA: 3.9</i>	August 2021 – May 2025 Poughkeepsie, NY
Budapest Semesters in Mathematics <i>Semester Study Abroad Program in Mathematics</i>	August 2024 – December 2024 Budapest, Hungary
University High School of Science and Engineering / University of Hartford <i>High School Diploma with Concurrent College Coursework</i>	August 2017 – June 2021 Hartford, CT
• Awarded Highest GPA, Most College Credits, Mathematics Scholar, and 4 Robotics Awards	

PUBLICATIONS & PREPRINTS

Beyond Distance and Density: Manifold Connectivity Shapes Similarity Judgments	2026
• Olivia A. Gatto*, Andrew J. Steindl *, Smita Krishnaswamy, Robert L. Goldstone	
• *Co-first authors. Accepted for talk at CogSci 2026	
Exploring the Manifold of Neural Networks Using Diffusion Geometry	2024
• Elliott Abel*, Andrew J. Steindl *, Selma Mazioud, Ellie Schueler, Folu Ogundipe, Ellen Zhang, Yvan Grinspan, Kristof Reimann, Peyton Crevasse, Dhananjay Bhaskar, Siddharth Viswanath, Yanlei Zhang, Tim GJ Rudner, Ian Adelstein, Smita Krishnaswamy	
• *Co-first authors. <i>arXiv preprint arXiv:2411.12626</i>	
Reeb Graphs and Towers: Multiscale Skeletons for Data	2025
• Andrew J. Steindl , Dhananjay Bhaskar, Smita Krishnaswamy	
• Accepted to <i>Topology, Algebra, and Geometry in Data Science (TAG-DS)</i> (non-archival)	
The Distribution of Genera of 2-Bridge Knots	2024
• Moshe Cohen, Abigail DiNardo, Adam M. Lowrance, Steven Raanes, Izabella M. Rivera, Andrew J. Steindl , Ella S. Wanebo (Alphabetical)	
• <i>Fundamenta Mathematicae</i> , 267(3):195–229	
• arXiv:2307.09399	
Distilling Language Models via Diffusion Geometry of Representations	Target: NeurIPS 2026
• Andrew J. Steindl , César Miguel Valdez Córdova, Folu Ogundipe, Ian Adelstein, Dhananjay Bhaskar, Smita Krishnaswamy	
• Target submission: NeurIPS 2026	
ReebBench: Discovering Geometry from High-Dimensional scRNA-seq Data	Target: NeurIPS 2026
• Andrew J. Steindl , João Felipe Rocha, Smita Krishnaswamy	
• Target submission: NeurIPS 2026 (Datasets & Benchmarks)	

TALKS

Beyond Distance and Density: Manifold Connectivity Shapes Similarity Judgments

- Cognitive Science Society (CogSci) July 2026 (Talk)

dReeb Graphs and Towers

- TAG-DS NeurIPS Workshop December 2025 (Poster)

Exploring the Landscape of Neural Networks Using Diffusion Geometry

- Jane Street Mathematics Conference July 2024 (Poster)
- Hudson River Mathematics Conference April 2025 (Talk)
- Deep Math Conference November 2025 (Poster)

Embedding Knots Via Learned Strategies of To Knot or Not to Knot

- Unknot V Conference July 2024 (Talk)
- Hudson River Mathematics Conference April 2024 (Talk)

Construction and Design of a Humanoid Robot

- Vassar URSI Symposium September 2023 (Poster)

The Distribution of Genera of 2-Bridge Knots

- Hudson River Mathematics Conference April 2023 (Talk)
- SUNY New Paltz Symposium for Undergraduate Math Research September 2022 (Talk)
- Vassar URSI Symposium September 2022 (Poster)

EXPERIENCE

SUMRY (Summer Undergraduate Mathematics Research at Yale)

Summer 2026

Mentor

New Haven, CT

- Mentor undergraduate students in geometric and computational mathematics research
- Guide projects, provide technical feedback, and support development of research presentations

Prison Mathematics Project

2025 – Present

Mentor

- Mentor incarcerated students in mathematics through written correspondence
- Provide feedback on problem sets and support mathematical development

Vassar College Mathematics Academic Internship

Spring 2025

Mathematics Academic Intern

Poughkeepsie, NY

- Hosted seven office hours per week to assist students with mathematics courses at Vassar

Summer Undergraduate Mathematics Research at Yale Fellowship

Summer 2024

Research Fellow

New Haven, CT

- Completed a 10-week mathematics REU on geometric manifold learning
- Produced a research paper

Autonomous Robotics Design Competition at Vassar College

Spring 2023

Student Designer and Builder

Poughkeepsie, NY

- Designed and built an autonomous robot to hunt and catch a robotic turtle as quickly as possible

Undergraduate Research Summer Institute at Vassar Fellowship

Research Fellow

Summer 2023
Poughkeepsie, NY

- Completed a 10-week robotics REU on humanoid robotics
- Produced a humanoid robot

Supervised Independent Study on Robotic Flapping Flight

Independent Researcher

Fall 2023
Poughkeepsie, NY

- Designed and built a biomimetic modular ornithopter to study the evolution of avian feathers

Undergraduate Research Summer Institute at Vassar Fellowship

Research Fellow

Summer 2022
Poughkeepsie, NY

- Completed a 10-week mathematics REU on knot theory
- Produced a research paper

AWARDS

Mary Evelyn Wells and Gertrude Smith Prize for excellence in mathematics or statistics

2025

- Awarded for outstanding academic achievement and demonstrated excellence in mathematics or statistics.

Yale GSAS Conference Travel Award

2025

- Funding to present research at an academic conference

Unknot V Travel Grant

2024

- Funding to travel and speak at the Unknot V conference

Thompson Bartlett Mathematics Fellowship

2024

- Funding to conduct mathematics research

Vassar College Film Festival – First Place Best Film

2023

- Awarded first place for a short film

COMMUNITY & OUTREACH

Mentor, SUMRY (Summer Undergraduate Math Research at Yale)

2026

- Mentoring undergraduate researchers in geometric and computational mathematics

Mentor, Prison Mathematics Project

2025 - Ongoing

- Teach and support mathematics education for incarcerated students

Hosted and Planned Vassar Integration Bee

2024

- Organized a department-wide competition bringing together faculty and students
- Designed event to foster mathematical community and engagement

Built the WebApp Flow Plane

2024

- Developed a web application to simulate and visualize 2D dynamical systems from user-defined differential equations
- Implemented tools for estimating fixed points and analyzing system behavior

Combat Robotics Initiative

2025

- Founded and developed a student robotics program, including arena construction and competition funding

Distributed PPE During COVID-19

2020

- Coordinated a 3D printing operation producing protective equipment for healthcare facilities
- Secured funding through the Southington Cultural Arts Commission (SoCCA) Grant and converted a vacant bar into a production site

INTERESTS

Representation & Geometry: geometric manifold learning, diffusion geometry, representation learning

Cognition & Learning: structure and dynamics of biological and machine representations, category learning, dynamical systems

Machine Learning: autoencoders, generative models, structure-aware learning

Mathematics: differential topology and geometry, geometry of sampled manifolds, knot theory

Scientific Computing: MATLAB, SnapPy, SageMath, LEAN

Embodied & Physical Systems: robotics, ROS 2, Micro-ROS, DepthAI

Fabrication & Prototyping: Fusion 360, SolidWorks, 3D printing